

AI vs Human Detection

By Group 14



Motivation

Motivation: Why Does This Matter?

- With the advancement of LLM, distinguishing between human-written and AI-generated text has become increasingly challenging

- Applications: Academic Integrity, Content Moderation, AI Transparency

RoadMap

catalog on all sections





01 Introduction

02 Datasets

- Controlled - Easy
- HC3 - Medium
- M4 - Hard

03 Baseline Models

- Baseline A - TF-IDF
- Baseline B - Signal Features
- Baseline C - Fine-tuned RoBERTa-base

04 Ensembles

- Single Layer Ensembles
- MultiHierarchial Ensemble

05 Evaluation

- Result
- Key Findings
- Error Analysis

06 Conclusion



Problem Definition & Motivation

Problem: Detecting AI-Generated Text **Binary classification**: given a piece of text, predict

$$y = 0 \text{ (Human-written)} \quad \textit{vs.} \quad y = 1 \text{ (AI-generated)}$$

Experiment Workflow

(1) Classical baselines: Baseline A(TF-IDF) + Baseline B(signal features)
(2) Transformer: fine-tune **RoBERTa-base** for deep semantic representations
(3) Ensemble: learned **voting machine** (neural net) combining all three models
(4) Evaluate *in-domain* and *cross-generator / cross-domain* generalization

Datasets --- Three Benchmarks



Each dataset tests a different level of generalization:

01

Controlled --- Easy (300 Human / 300 AI)

- Single generator: GPT-4o-mini
- Limited domain variety: essay, Wikipedia, news

02

HC3 --- Medium (2000 Human / 2000 AI)

- Q&A conversational responses to the same open-ended questions
- Single generator: ChatGPT based on GPT3.5
- Three domains: Wikipedia, Open-domain Question Answering, and Reddit(conversational)

03

M4 --- Hard (7185 Human / 7185 AI)

- 5 generators: ChatGPT, GPT-3 (davinci003), Cohere, Dolly, BLOOMZ
- 5 domains: arXiv(academic), PeerRead(very academic), Wikipedia, Reddit/ELI5(conversational), WikiHow(instructional)

Hypothesis: *Transformer-based models will generalize better than lexical or signal-only baselines as dataset complexity increases.*





Baseline Model

Baseline Models

01 Baseline A ---- TF-IDF + Logistic Regression

- Represent each text as a **unigram + bigram TF-IDF** vector (top 20,000 highest-weights features, stop-words removed)
- Train a **class-balanced Logistic Regression**
- Captures *word-choice patterns* that differ between human and AI

02 Baseline B ---- Signal Features + Logistic Regression

Six hand-crafted stylometric features per text, extracted with GPT-2:

- **Perplexity** (GPT-2) --- how "surprising" the text is overall
- **Burstiness** --- variation in sentence-level fluency
- & Sentence-length std, type-token ratio, punctuation ratio, word count

Features are **StandardScaled** before Logistic Regression.
Inductive Bias: AI text is *uniformly fluent*; human text *varies more*.



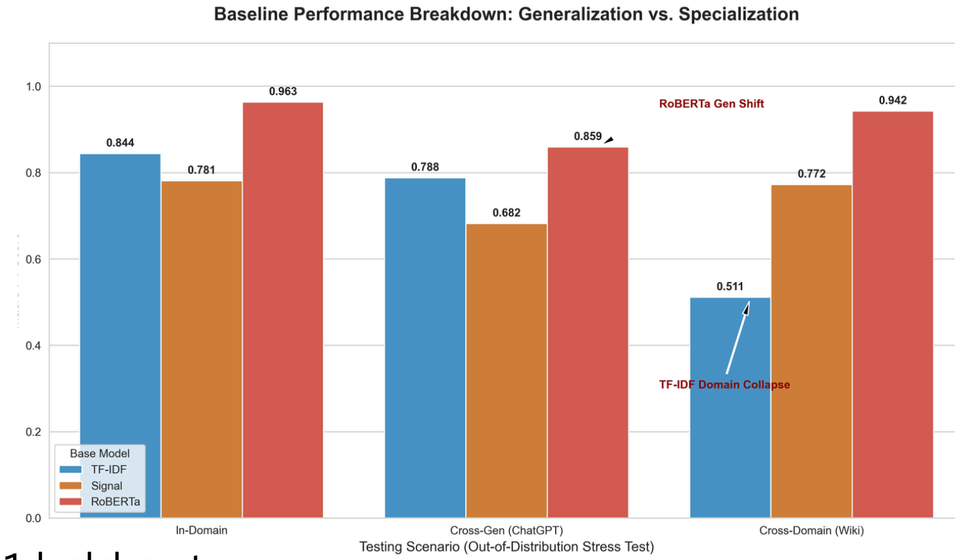
In-Domain [Train(70) / Dev(15) / Test(15) from the *same* dataset]:

- Signal Features (B) is strongest on Controlled and HC3 [global stylistic cues remain], while TF-IDF (A) is strongest on heterogeneous M4 in-domain [lexical artifacts remain exploitable].

Baseline Models

03 Baseline C ---- Fine-tuned RoBERTa-base

- A pre-trained RoBERTa-base transformer (125M parameters).
- Full-parameter fine-tuning on each training set (max sequence length: 512 tokens).
- Captures deep semantic and contextual representations that identify subtle neural generation signatures.



Can baselines handle unseen generators / domains? (Macro-F1)

Cross-Generator

Train on 4 generators; test on 1

Held-out	TF-IDF (A)	Signal (B)
chatgpt	0.788	0.682
davinci003	0.812	0.701
cohere	0.794	0.731
dolly	0.768	0.654
bloomz	0.759	0.567

Cross-Domain

Train on 4 domains; test on 1 held-out

Held-out	TF-IDF (A)	Signal (B)
wikipedia	0.512	0.773
reddit_eli5	0.692	0.744
wikihow	0.677	0.700
arxiv	0.634	0.719
peerread	0.859	0.726

Takeaway:

Under **generator shift**, TF-IDF (A) is consistently better; Under **domain shift**, Signal Features (B) is often more robust (**except** PeerRead, where TF-IDF remains best).

Trade-off between lexical memorization and global, domain-robust cues => Fine-tuning RoBERTa and Ensembles

Rationale for Ensemble Model

Each baseline captures a different view of the text:
lexical, global-features, and contextual

Instead of choosing one detector,
we combine their predictions to improve robustness









Ensemble/Fusion Model



Ensemble

01 Single Level Ensemble

- Naive Average
- SimpleStacking
- SimpleAwareMLPGate

02 Multi-Hierarchial Ensemble

- Double Layer ensemble

Single Layer Ensemble

01 Naive Average

- Take the average

```
test_naive = np.mean([test_t, test_s, test_r], axis=0)
```

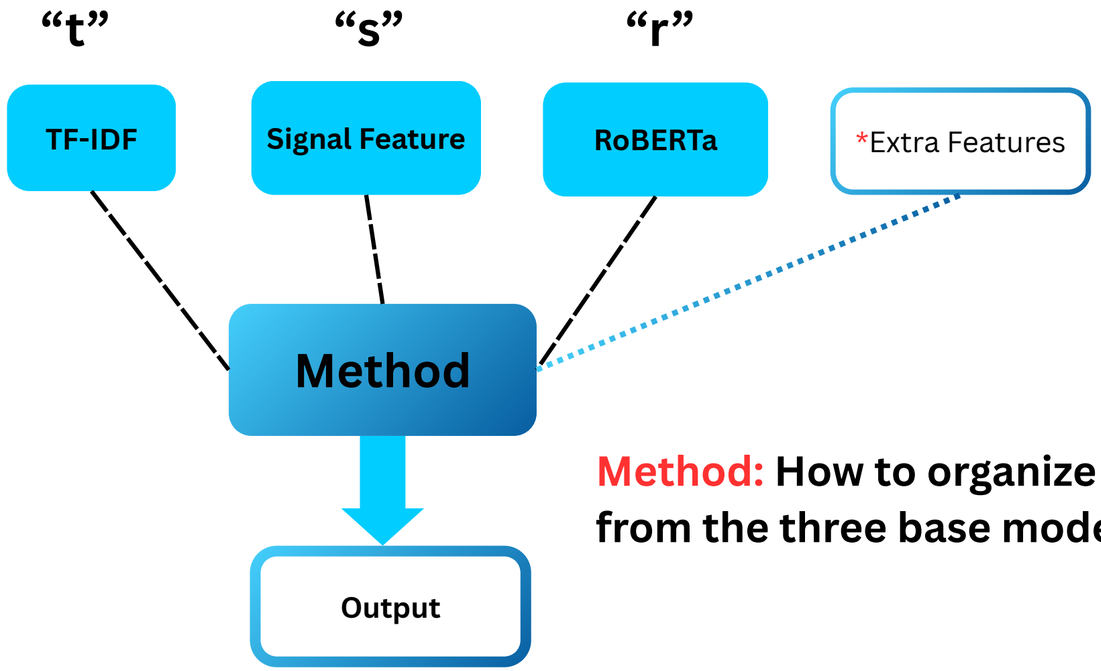
02 Simple Stacking

- Stack the **X** sets(development and test)
- Fit **Logistic Regression** with stacked X development
- Predict with the fitted **Logistic Regression**

```
stacker = LogisticRegression(random_state=42)
stacker.fit(X_dev_stack, dev_labels)
test_stacking = stacker.predict_proba(X_test_stack)[:, 1]
```

03 *Signal Aware MLP Gate

- Build **X** sets(development and test) with splitted sets for each model and **Extra features**
- Train **MLP Gate** with the development X,Y sets
- Test with the X test set on the **MLP Gate** trained



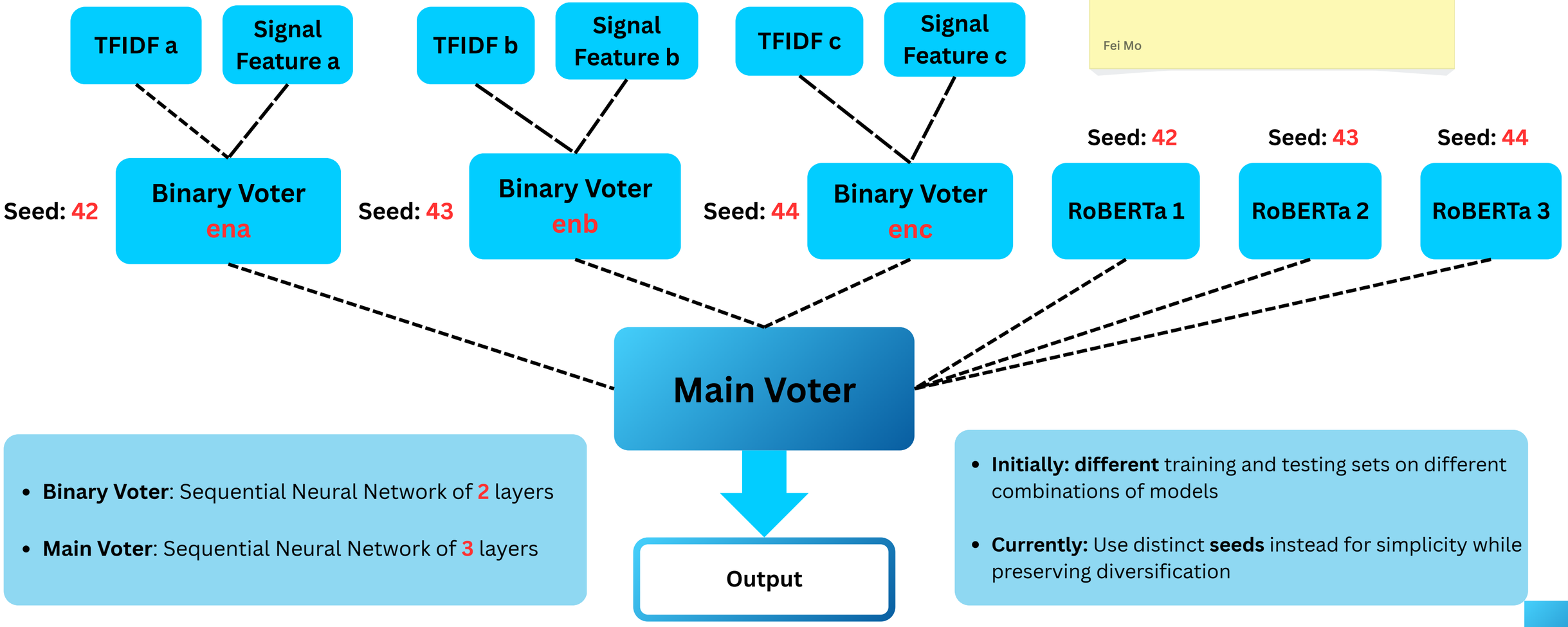
Method: How to organize results from the three base models ?

Double Layer Ensemble

Models:

- 9 small individual models
- 3 binary ensembles
- 1 main meta ensemble

Fei Mo



- **Binary Voter**: Sequential Neural Network of 2 layers
- **Main Voter**: Sequential Neural Network of 3 layers

- **Initially: different** training and testing sets on different combinations of models
- **Currently:** Use distinct **seeds** instead for simplicity while preserving diversification





Evaluation

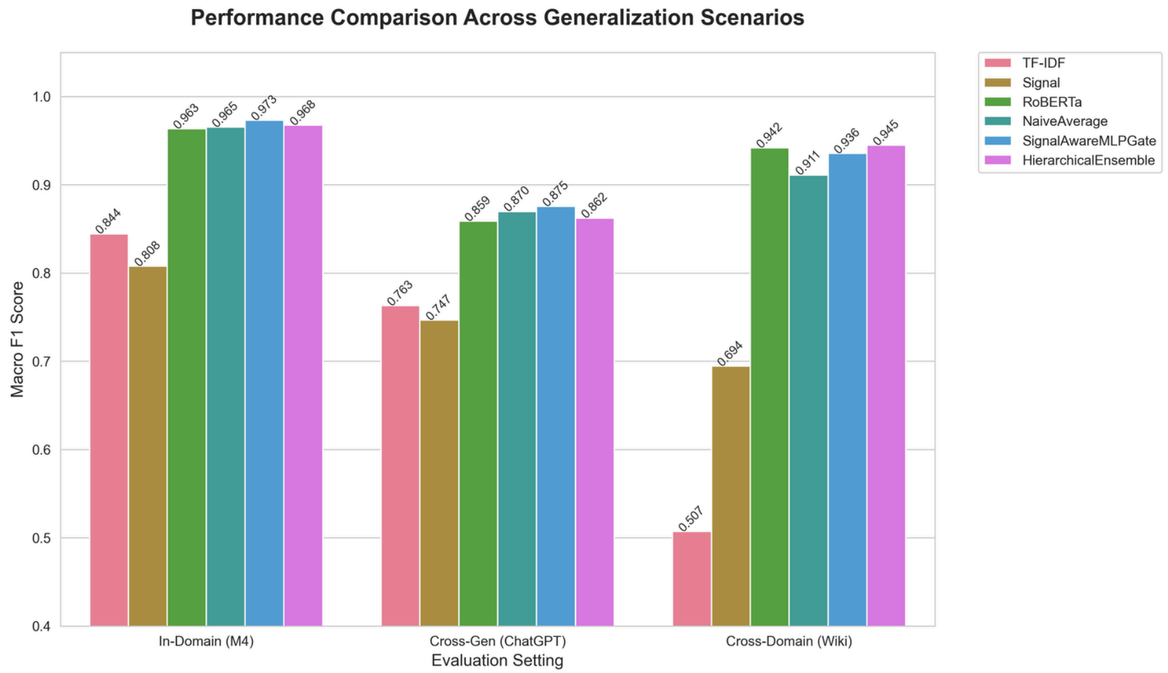
Evaluation - Main Results

Metrics: Accuracy, Precision, Recall, AI-class F1, Macro F1, ROC-AUC

Primary metric: Macro F1

Main results

Setting	Method	AI F1	Macro F1
M4 → M4	TF-IDF	0.839	0.844
M4 → M4	Signal	0.801	0.808
M4 → M4	RoBERTa	0.964	0.963
M4 → M4	Signal-Aware MLP Gate	**0.973**	**0.973**
Held-out ChatGPT	TF-IDF	0.605	0.763
Held-out ChatGPT	Signal	0.596	0.747
Held-out ChatGPT	RoBERTa	0.764	0.859
Held-out ChatGPT	Signal-Aware MLP Gate	**0.791**	**0.875**
Cross-Domain Wiki	TF-IDF	0.645	0.507
Cross-Domain Wiki	Signal	0.716	0.694
Cross-Domain Wiki	RoBERTa	0.945	0.942
Cross-Domain Wiki	Hierarchical Ensemble	**0.947**	**0.945**

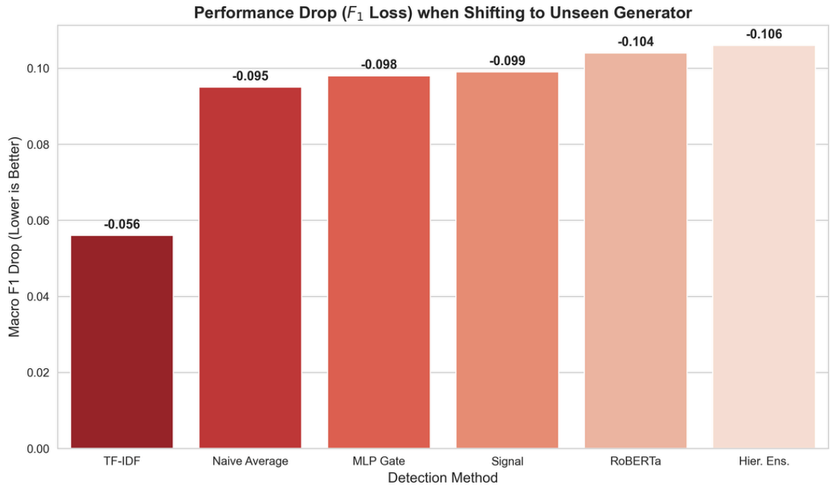


Interpretation

- Signal features are not the best standalone detector, but they provide complementary information for fusion.
- The Signal-Aware MLP Gate improves Held-out ChatGPT Macro F1 from 0.859 to 0.875, showing better robustness to unseen generators.

Evaluation - Robustness

Robustness Analysis: Performance Drop under Shift



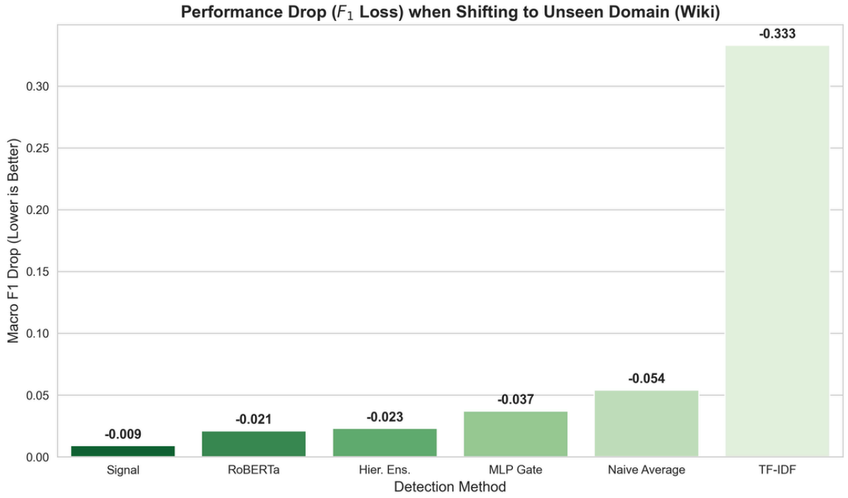


Table 5: Evaluation settings and robustness dimensions.

Setting	What changes?	Purpose
M4 → M4	No distribution shift	In-domain upper bound
Held-out ChatGPT	Generator changes	Cross-generator robustness
Cross-Domain Wiki	Domain changes	Cross-domain robustness

- Generator shift challenges all models.
  Even RoBERTa drops, suggesting **detectors** still learn **generator-specific** cues
- **Domain** shift mainly hurts **TF-IDF**.
  Lexical patterns are **fragile** when topic vocabulary changes
- **Signal features** are relatively **stable** under Wiki domain shift
- **Fusion** is useful because the models fail differently:
  TF-IDF sees **words**, Signal sees **style**, RoBERTa sees **context**

Evaluation - Signal Intepretation

Signal Feature Interpretation and Rationales

Feature	Captures	Why it helps
Perplexity	overall predictability	AI text is often more fluent and easier for LM to predict
Avg Log Prob	model confidence	Reflects how confidently the LM generates each token
Burstiness (LM)	token-level loss variation	Human writing varies more locally; AI is often smoother
Burstiness (Sent)	sentence-length variation	Humans mix long and short sentences more naturally
Type-Token Ratio	lexical diversity	AI text may repeat vocabulary more in long passages
Punctuation Ratio	writing rhythm / formatting	Punctuation usage reveals stylistic regularity

The Role of Signal Features in Fusion

- Signal features provide **non-lexical** and **non-semantic** cues that complement RoBERTa.
- While RoBERTa captures contextual meaning, signal features capture the **statistical signature** of the generation process.
- They are especially useful as **reliability** cues when the neural detector becomes less stable under generator or domain shift.

Signal features do not replace RoBERTa —
they complement it by adding stable statistical evidence.



CONCLUSION

- **3 Base Models < Double Layer Ensembling <= *3 Single Layer Ensembles**
  - Single Layer Ensemble - Simple Stacking & Naive Average - suits our task better
- **Limitation:**
  - The models for Double Layer Ensemble are not as diverse - may affect the performance

Thank you!

Comments / Feedback ?

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